

Project Report: Quiz Master V2

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Description

Quiz Master - V2 is a multi-user application that serves as an exam preparation site. The platform is designed with two distinct roles: an Admin (Quiz Master) and multiple Users. The administrator is responsible for managing the platform's content, including subjects, chapters, quizzes, and questions, as well as overseeing user activity. Users can register, log in, attempt quizzes on various subjects, and track their performance and scores.

Technologies used

The project is built using a combination of modern frameworks and technologies to ensure robustness, scalability, and an interactive user experience.

Backend: Flask is used as the backend framework for handling API routing and application logic.

- Flask-Login: Manages user authentication and session control.
- Flask-SQLAlchemy: Serves as the Object-Relational Mapper (ORM) for seamless database interactions.
- Werkzeug Security: Ensures secure password management by hashing and verifying passwords.

Frontend: VueJS is utilized for building a dynamic and responsive user interface. Bootstrap is used for HTML generation and styling.

Database: SQLite is the relational database for all data storage.

Caching & Background Jobs: Redis is implemented for caching, while Redis and Celery are used to manage asynchronous backend jobs, such as sending reports and reminders.

DB Schema Design

User Table: Stores information for registered users.

- **id:** INTEGER (Primary Key) - A unique identifier for each user.
- **email:** STRING (Unique) - User's email, used as a username.
- **password:** STRING - Hashed password for secure authentication.
- **full_name:** STRING - The full name of the user.
- **qualification:** STRING - The user's educational qualification.
- **dob:** DATE - The user's date of birth.
- **is_admin:** BOOLEAN - A flag to differentiate between an admin and a regular user.

Subject Table: Contains details about the subjects available for quizzes.

- **id:** INTEGER (Primary Key) - A unique identifier for each subject.

- **name:** STRING - The name of the subject (e.g., Mathematics).
- **description:** TEXT - A brief overview of the subject.

Chapter Table: Holds information about chapters within each subject.

- **id:** INTEGER (Primary Key) - A unique identifier for each chapter.
- **subject_id:** INTEGER (Foreign Key) - Links to the **Subject** table.
- **name:** STRING - The name of the chapter (e.g., Algebra).
- **description:** TEXT - A brief description of the chapter's content.

Quiz Table: Defines the attributes of each quiz.

- **id:** INTEGER (Primary Key) - A unique identifier for each quiz.
- **chapter_id:** INTEGER (Foreign Key) - Links to the **Chapter** table.
- **date:** DATE - The date when the quiz is available.
- **time:** TIME - The duration allotted for the quiz in HH:MM format.
- **remarks:** STRING - Any additional notes or comments about the quiz.

Question Table: Stores the questions for each quiz.

- **id:** INTEGER (Primary Key) - A unique identifier for each question.
- **quiz_id:** INTEGER (Foreign Key) - Links to the **Quiz** table.
- **question_statement:** STRING - The text of the multiple-choice question.
- **option1, option2, option3, option4:** STRING - The four answer options for the question.
- **correct_option:** INTEGER - The index (1-4) representing the correct answer.
- **marks:** INTEGER - The weightage or marks for the question.

Score Table: Records the performance of users in each quiz they attempt.

- **id:** INTEGER (Primary Key) - A unique identifier for the score entry.
- **quiz_id:** INTEGER (Foreign Key) - Links to the **Quiz** table.
- **user_id:** INTEGER (Foreign Key) - Links to the **User** table.
- **total_scored:** INTEGER - The total score the user achieved in the quiz.
- **time_stamp_of_attempt:** DATETIME - The exact date and time the quiz was attempted.

Architecture and Features

The project follows a structured architecture to separate concerns and enhance maintainability:

- **Controllers (app.py):** Flask routes that handle incoming HTTP requests and manage the application's logic.
- **Models:** Defines the SQLAlchemy database tables and their relationships.
- **Templates (templates/):** Contains the main HTML file rendered by Jinja2, which serves as the entry point for the VueJS application.
- **Static Files (static/):** Stores the compiled VueJS assets (JavaScript, CSS) and other static files like images.

Features Implemented

Core Functionalities

- **Role-Based Access:** Distinct login and dashboard functionalities for Admin and User roles.
- **Admin Dashboard:** A central hub for the admin to perform the following actions:
 - Create, edit, and delete subjects, chapters, quizzes, and questions.
 - View and search all registered users.
 - View summary charts, such as quiz attempt distributions, for data visualization.
- **User Dashboard & Quiz Taking:**
 - Allows users to register, log in, and log out.
 - Attempt any available quiz, which includes a timer to enforce time limits.
 - View scores from all previously attempted quizzes.
 - See personalized summary charts, like average scores per quiz.

Backend Jobs (Asynchronous Tasks)

- **Daily Reminders:** A scheduled job runs daily to send reminders (via email or other webhooks) to users who have been inactive or when a new relevant quiz is created.
- **Monthly Activity Report:** On the first day of each month, an automated job generates and emails a personalized HTML activity report to each user, detailing quizzes taken, scores, and performance analysis for the previous month.
- **CSV Export:** Users can trigger an asynchronous job from their dashboard to export the details of their completed quizzes (e.g., quiz ID, date, score) into a CSV file. The user is notified once the file is ready for download.

Video

[MAD 2 Demo](#)